

Requirements Rough Draft – C&C

Date:

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Researcher(s):

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Objective of research:

Begin thinking about what requirements we should have for the ROV and create a few for practice that we can discuss later as a group

Notes:

Requirement 1: The ROV's microcontroller shall use Python coding for all computations

Rationale: Our team is currently more comfortable with Python and has more experience with that coding language

Needs this is based on: C&C 1.1, C&C 1.3

Issues/concerns/comments: What coding language we decide to use is still up for debate

Requirement 2: The ROV shall use one Arduino Pro Mini/Nano as a microcontroller

Rationale: Small microcontroller to easily fit inside the ROV and develop code on

Needs this is based on: C&C 1.1, C&C 1.1.1

Issues/concerns/comments: Arduinos usually use C++ coding which could interfere with requirement 1, using Python might be possible but could be more work. Most likely would have to use a different microcontroller if we fully decide on Python.

Requirement 3: The ROV shall use one camera to give the pilot an understanding of where the ROV is

Rationale: Using one camera would be the most cost-efficient, allows the pilot to see in front of the ROV. In the 2024 Pioneer competition, the Merknights ROV used only one camera in order to stay under their budget, so it is possible to operate the ROV with only one camera.

Needs this is based on: C&C 1.4

Issues/concerns/comments: Number of cameras we should use is still up for debate, could depend on budget. Also, we could need more cameras depending on what kind of tasks the ROV has to accomplish.

Sources:

https://drive.google.com/file/d/1Wda_ZJjK9ycHMS2GFzKe2RmPIOyOglU9/view?usp=sharing

Requirement 4: The ROV shall use **XX** inches of wire to send signals between devices

Rationale: Devices on the ROV will need to send signals to each other through wire

Needs this is based on: C&C 1.2, C&C 1.2.1, C&C 1.2.2, C&C 1.2.3

Issues/concerns/comments: Unsure how much wire is necessary at this time, unsure what type of wire or thickness at this time

Sources:

MATE ROV Mission Requirements page